Operating Systems - Chapter 5: CPU Scheduling

Demonstration document - fictional data

5.1 Why Scheduling Exists

A CPU can execute one thread per core at a time, while dozens of processes are ready to run. The scheduler decides which ready process runs next. Good scheduling maximizes CPU utilization and throughput while minimizing turnaround time, waiting time, and response time.

5.2 First-Come First-Served (FCFS)

FCFS runs processes in arrival order without preemption. It is simple and fair in order of arrival but suffers from the convoy effect: one long CPU-bound process delays every process behind it, inflating average waiting time dramatically.

5.3 Shortest Job First (SJF)

SJF selects the process with the smallest next CPU burst. It is provably optimal for average waiting time, but burst lengths must be predicted (commonly with an exponential average of previous bursts). The preemptive variant is Shortest Remaining Time First (SRTF).

5.4 Round Robin (RR)

Round Robin gives each process a fixed time quantum, typically 10 to 100 milliseconds, then preempts it to the back of the ready queue. Small quanta improve response time but increase context-switch overhead; a quantum should be large relative to the switch cost.

5.5 Priority and Multilevel Feedback Queues

Priority scheduling runs the highest-priority ready process and risks starvation of low priorities, mitigated by aging. Multilevel feedback queues combine ideas: new processes start in a high-priority queue with a short quantum and are demoted if they use full quanta, approximating SJF without predictions.

5.6 Key Formulas

Turnaround time = completion time - arrival time. Waiting time = turnaround time - burst time. CPU utilization = busy time / total time. In RR with n processes and quantum q, no process waits longer than (n - 1) x q before its next turn.